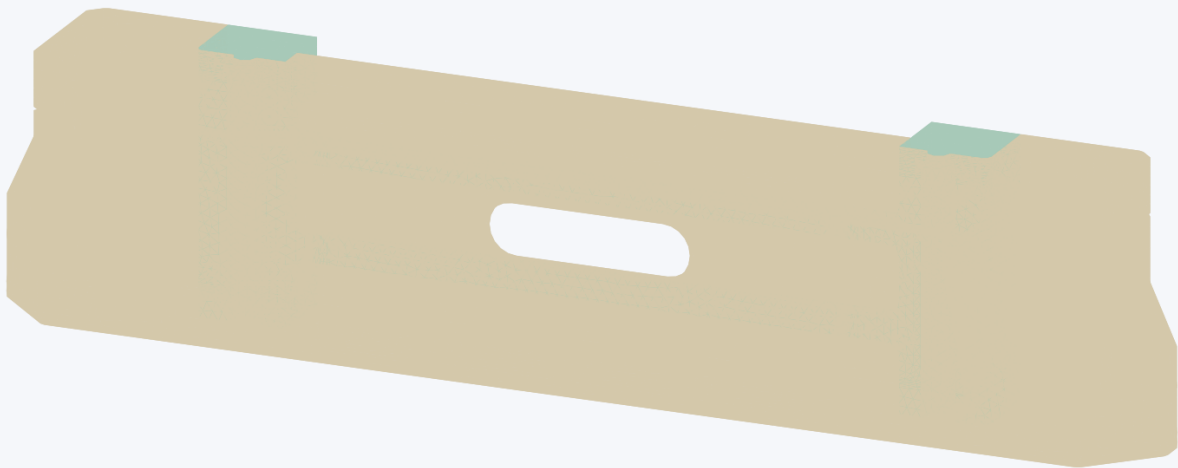


Selected-run evidence status: failed

Geometry

Undeformed geometry

Exterior shape from the selected mesh



Mesh-derived exterior • source geometry unchanged

Undeformed geometry from selected mesh. Exterior shape derived from the selected run's mesh; this is not a CAD render.

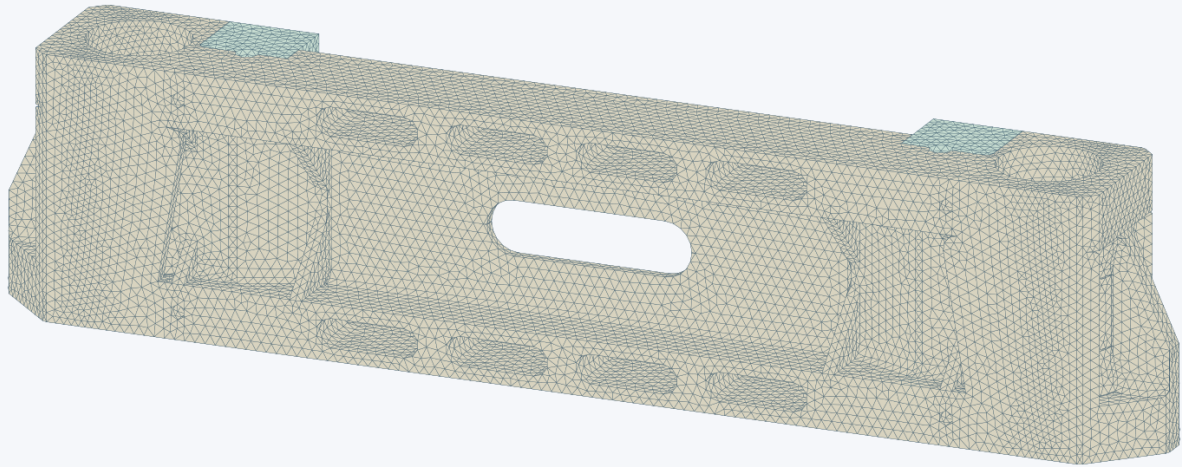
Mesh

Mesh overview and local detail

Mesh nodes	35031	
Tetrahedral elements	130760	
Meshed parts	2	
Nominal element size	3.0	mm
Minimum mean-ratio quality	0.0113	0-1

Mesh overview

Undeformed exterior mesh

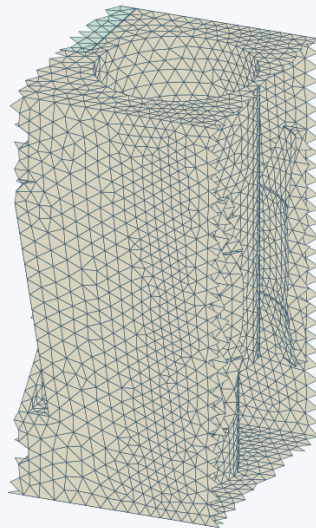


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Pad-root refinement detail

Pad-root region



Exterior mesh faces • undeformed geometry

Pad-root refinement detail. Positive pad-root region, $X = 112\text{--}150\text{ mm}$. Mesh metadata records a 0.65 target factor (1.95 mm local target).

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	0.2	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.01	ms
Startup mass-scaling DT2MS	-3e-07	s
Motor mass	0.1616	kg
Moving mass	0.6493825	kg
Pusher mass	0.4877825	kg
Initial kinetic energy	811.7281	J

Recorded material, contact and boundary assumptions

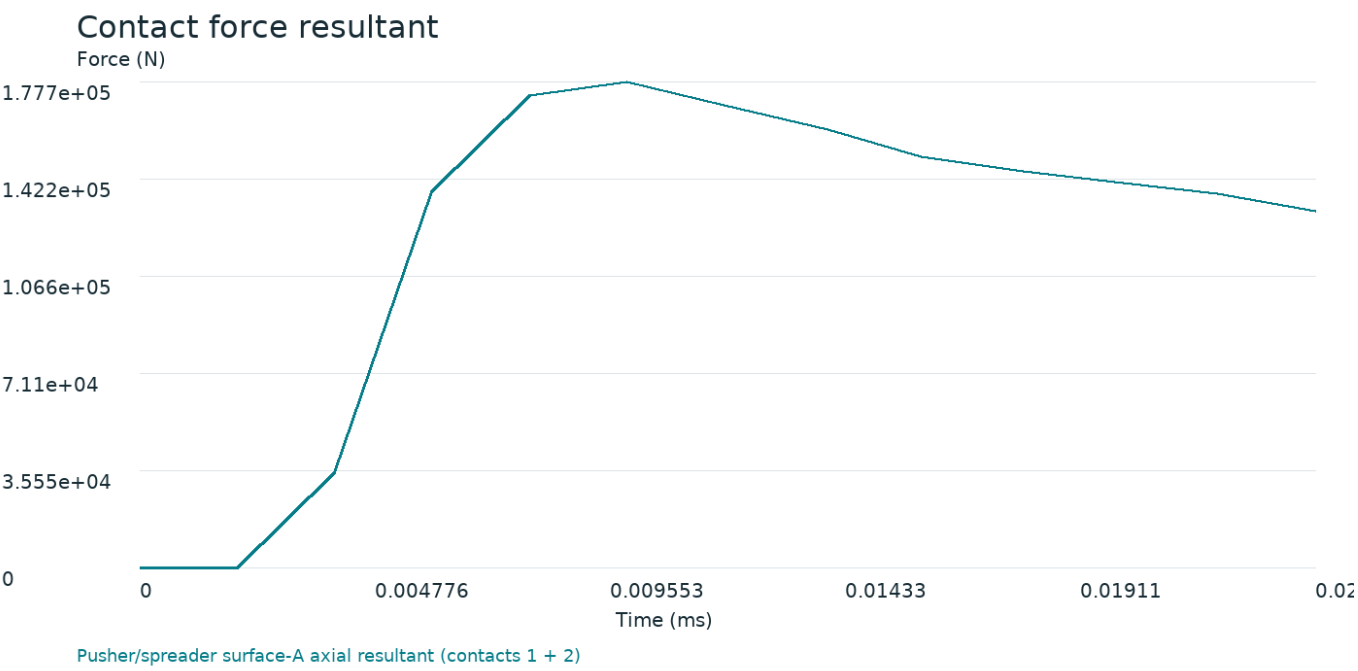
- Body and bolted cable cap flexible; four local perfectly clamped bolt collars. No bolt preload, slip or failure.
- Motor loaded upper-bound inertia distributed over actual bore nodes; no motor housing stiffness or exact inertia tensor.
- PA12-CF10 elastic isotropic proxy $E=1806.6\text{MPa}$ $\nu=.35$ density 1060; print +Z motor axes. No strength/failure claims.
- No fracture, erosion, motor thrust, carrier, released endcap or invented 35g hardware allowance.
- Rigid 8mm aluminum spreaders initially stationary; fixed x,y and rotations, free z without travel limit.
- Pusher has no external SPC constraints: all stopping reaction passes through transient contact.

- Foam rear z fixed, x/y free: frictionless fixed backing; sides unconfined.
- MAT057 synthetic loading and unloading; HU=.05 BETA=0 DAMP=.1 SHAPE=1 FAIL=0. Not a product material model.
- Generic foam anchored at one catalog point; remaining curve chosen for nonbottoming comparative stop.
- Startup-only local mass scaling requested; added mass and inertia error MUST be assessed from solver output.
- Mesh, timestep, energy and momentum require independent assessment.

Solve

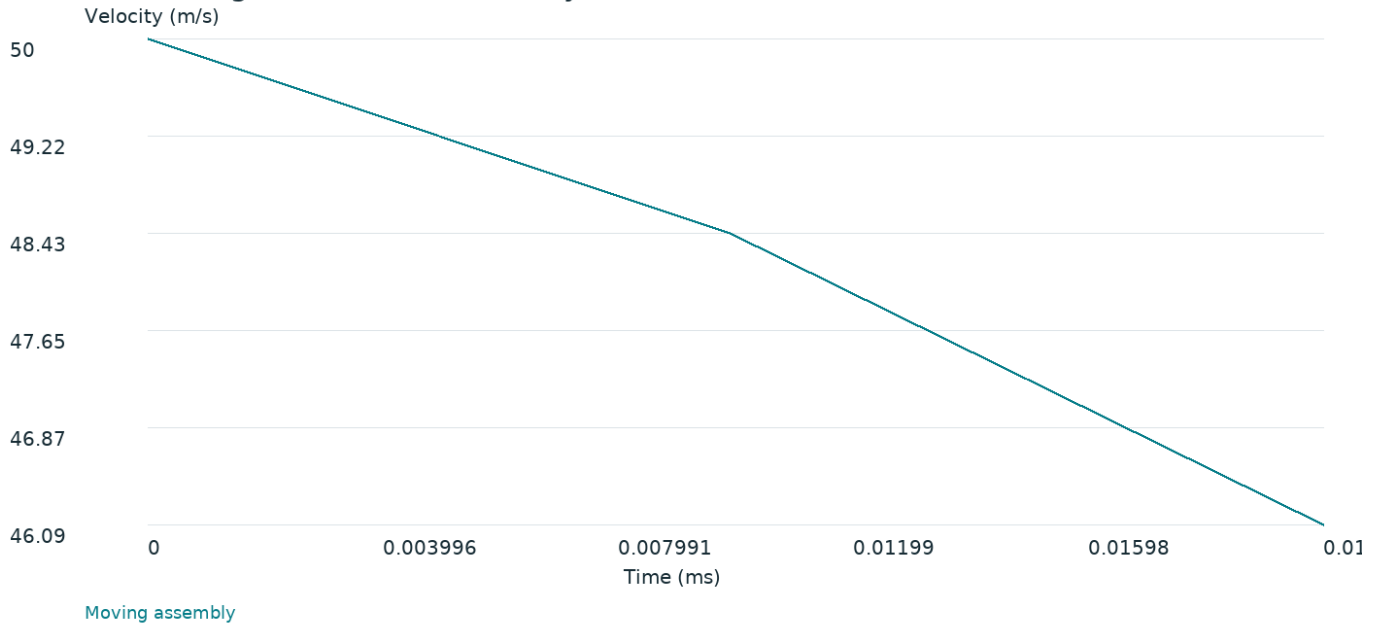
Recorded run status	failed
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Separate solve charts



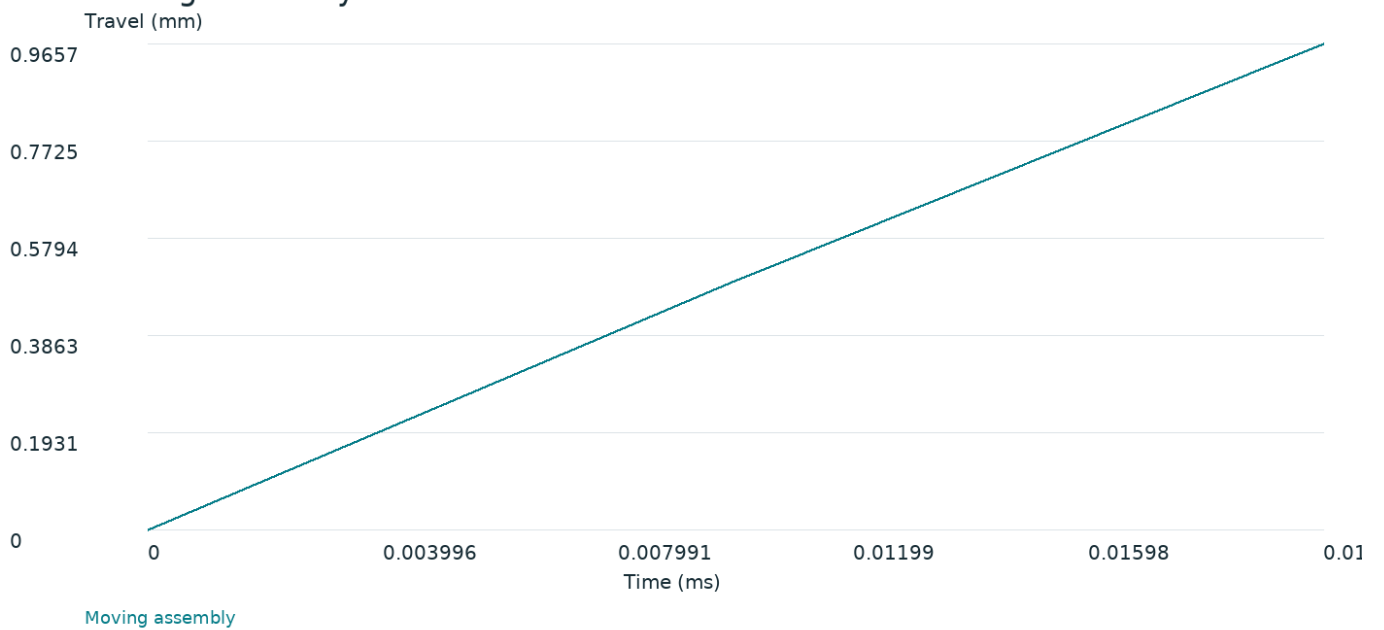
Contact force resultant. [projects/compliant_stop_340/cases/diagnostic-3mm-ms-v1/assessment/contact_histories.csv](#)

Mass-weighted forward velocity



Mass-weighted forward velocity. projects/compliant_stop_340/cases/diagnostic-3mm-ms-v1/assessment/time_histories.csv

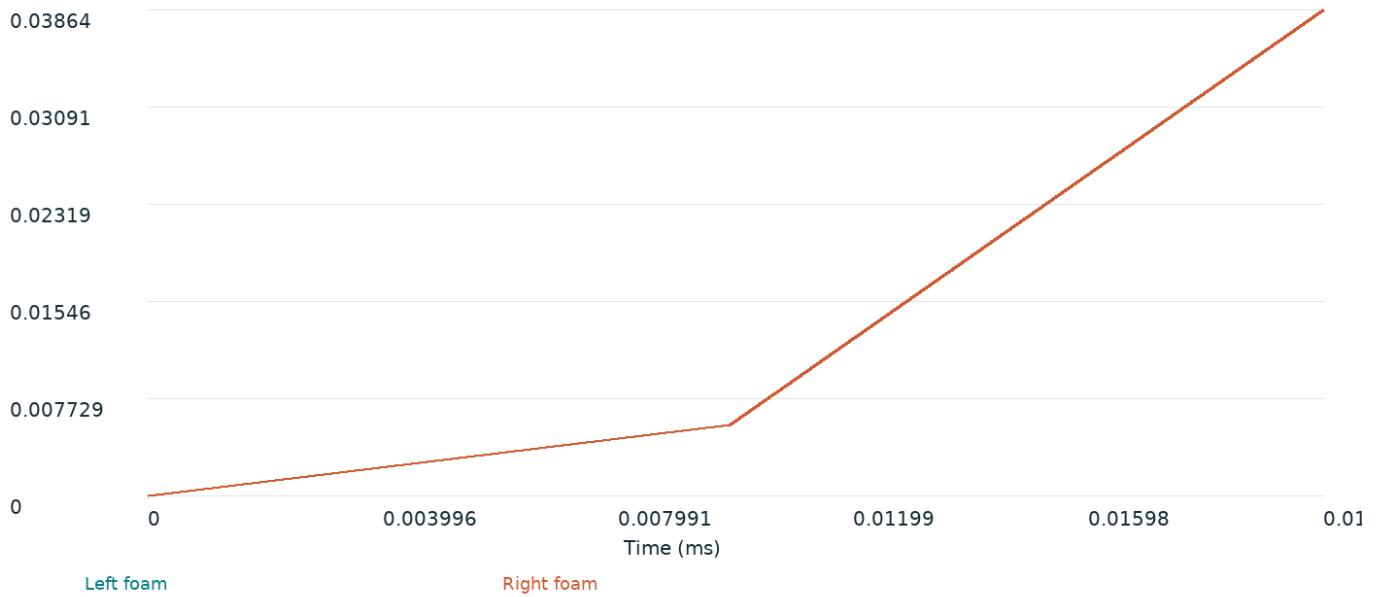
Moving assembly travel



Moving assembly travel. projects/compliant_stop_340/cases/diagnostic-3mm-ms-v1/assessment/time_histories.csv

Foam compression

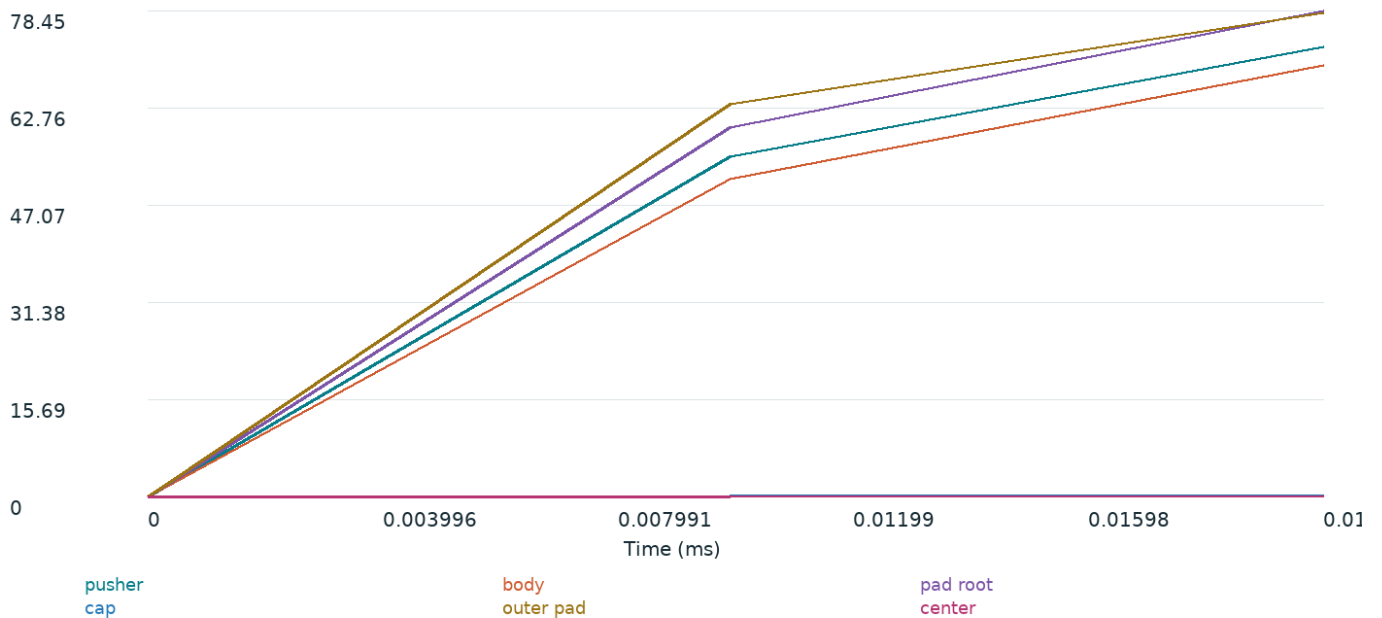
Compression (%)



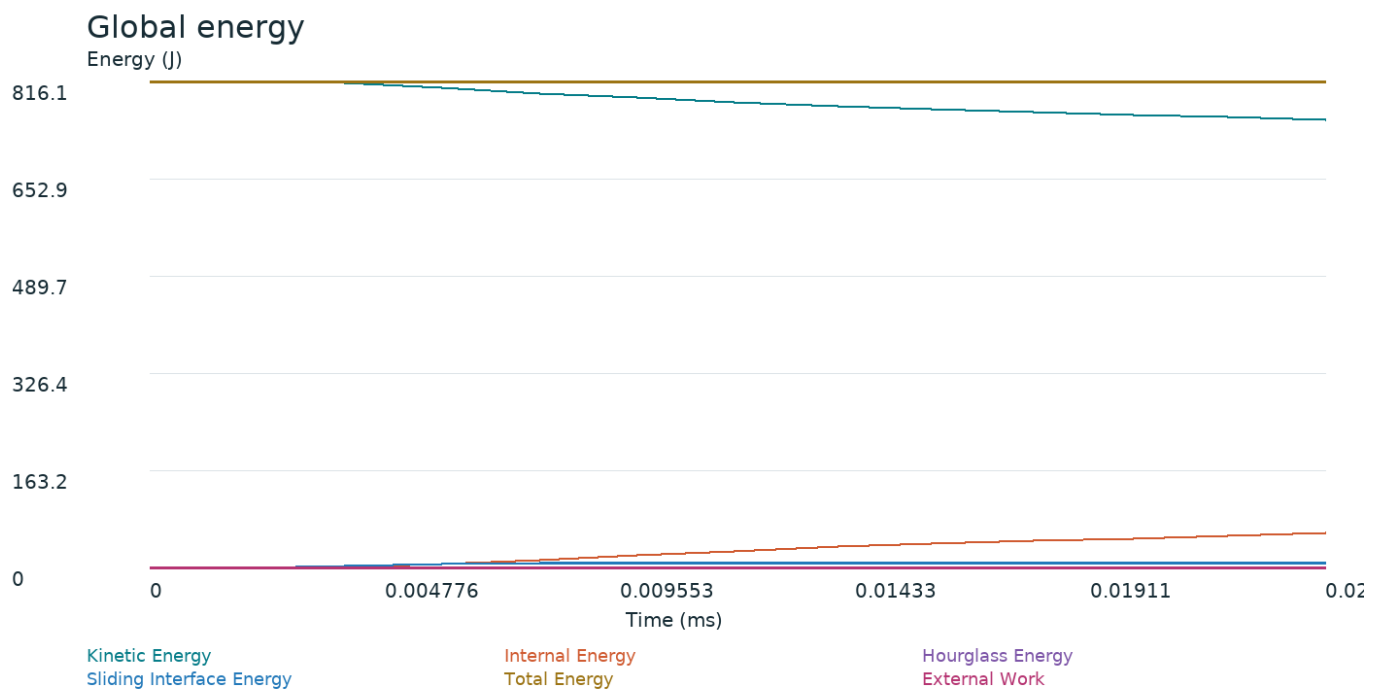
Foam compression. projects/compliant_stop_340/cases/diagnostic-3mm-ms-v1/assessment/time_histories.csv

Regional P99 von Mises stress

Stress (MPa)



Regional P99 von Mises stress. projects/compliant_stop_340/cases/diagnostic-3mm-ms-v1/assessment/stress_histories.csv



Global energy. projects/compliant_stop_340/cases/diagnostic-3mm-ms-v1/assessment/global_energy_histories.csv

Results

Run 014 - impact evidence

Evidence status: failed. This report describes only extracted evidence from the selected run. Numerical checks alone do not establish physical qualification.

Geometry and model setup

Recorded model item	Value
kind	impact
pusher kg	0.487782
body kg	0.415659
cable cap kg	0.0721237
motors kg	0.1616
moving kg	0.649382
initial ke J	811.728
spreader each kg	0.135871

Solved mesh

Recorded mesh source: mesh_3mm. Mesh and timestep convergence are not established by these histories.

Motion and stopping event

Signed forward velocity, travel and foam compression are measured histories. Negative forward velocity denotes rebound; a partial history cannot establish the complete stopping event.

Measured result	Value	Unit
Peak stopping force	177745	N
Maximum travel	0.965687	mm
Peak regional P99 (pad root)	78.4493	MPa
Final forward velocity	46.0855	m/s

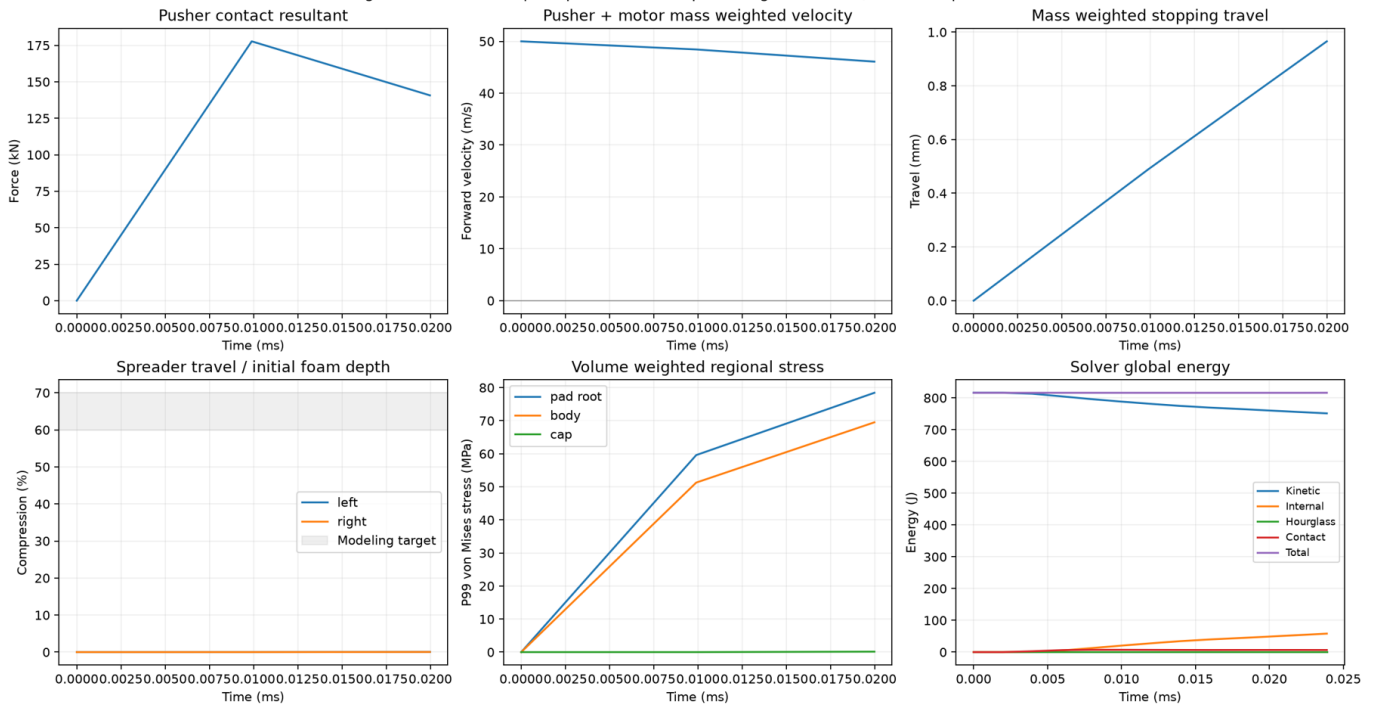
Regional stress and energy

Regional P99 values are elastic demand measures across saved states, not printed-part strength predictions. Force peaks and percentile peaks can occur at different times.

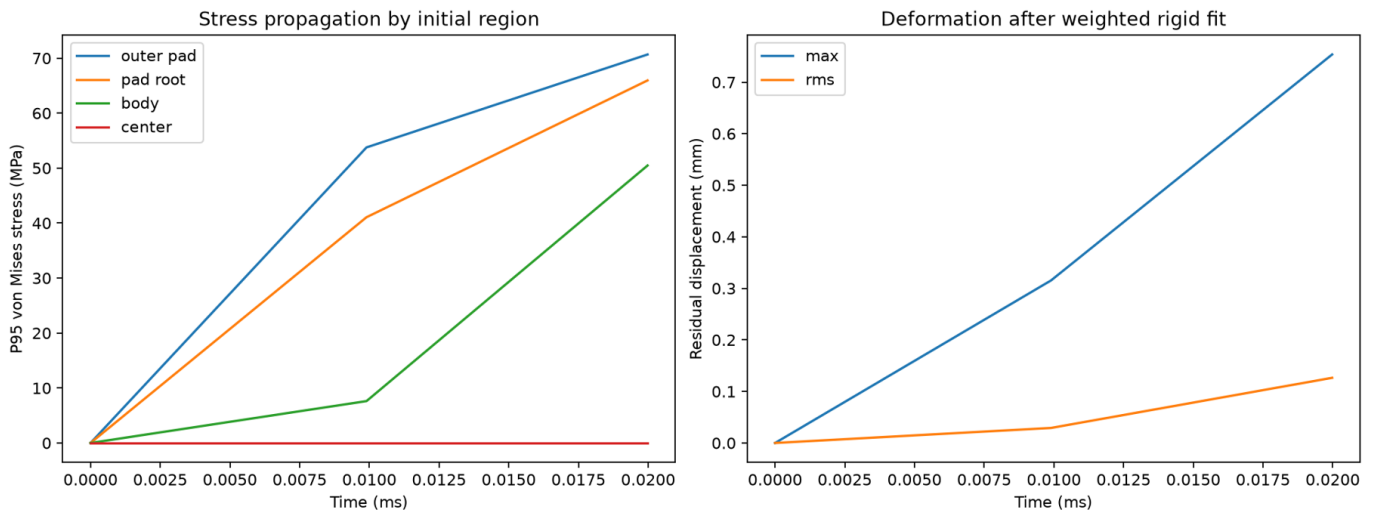
Numerical checks and qualification

Extracted numerical check	Evidence
normal termination	false
requested time reached	false
binary matches nodout	true
moving mass consistent	true
initial energy error below 0p5pct	false
energy balance error below 5pct	true
hourglass below 5pct initial energy	true
absolute contact energy below 5pct initial energy	true
contact impulse momentum error below 5pct	true
no axial spc on moving or guide nodes	true
rigid guides leave only axial translation free	true
no prescribed motion	true
no detected deletion	true
positive sampled foam jacobians	true
first stop captured	false
foam compression below 80pct	true
mass scaling disabled or quantified below 0p1pct moving mass	false

Images, animations and source evidence



histories.png



stress-propagation-deformation.png

- `contact_histories.csv`
- `global_energy_histories.csv`
- `metrics.json`
- `part_energy_histories.csv`
- `spc_histories.csv`
- `stress_histories.csv`
- `time_histories.csv`

Generated from selected-run extracted evidence; no authored report is available. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.